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PALAEOMAGNETISM OF THE GAWLER RANGE VOLCANICS AND IMPLICATIONS FOR THE GENESIS OF THE MIDDLEBACK HEMATITE OREBODIES

By F. H. CHAMALAUN & C. E. DEMPSEY

(With 1 Table and 5 Text-Figures)

(MS received 24 July 1978)

ABSTRACT

A new Precambrian palaeomagnetic pole has been obtained for the Gawler Craton, South Australia. The pole is based on results obtained from the accurately dated Gawler Range Volcanics (1525 ± 15 m.y.) and is situated 230°E 60.4°N with $\alpha_{95} = 6.2°$. The pole is in excellent agreement with poles previously determined from the negatively magnetized ore of Iron Monarch and the negatively magnetized dykes that intrude the Gawler Craton. The results suggest that the iron ore-forming process is closely related to igneous activity and supports the hypothesis of a hypogene origin for the ore.

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INTRODUCTION

The theory of plate tectonics has successfully combined palaeomagnetically determined polarwander curves and marine geophysical data into a coherent description of continental motions during the Phanerozoic (McElhinny, 1973). Extrapolation of these concepts to the Precambrian is of considerable interest for an understanding of the development of intracratonic mobile belts and the general tectonic regime prevailing during earliest history of the Earth's crust (Rutland, 1973; Sutton & Watson, 1974; Smith, 1976; McElhinny & McWilliams, 1977). Comparison of palaeomagnetically-determined polarwander curves from different cratons within the same continent may reveal the extent and timing of relative motions between cratons. At the same time polarwander curves may be used as a magnetostratigraphical tool to determine the age or relative stratigraphical position of formations that cannot otherwise be dated. Progress in Precambrian palaeomagnetism is, however, likely to be slow because of the much longer rock record involved compared to that of the Phanerozoic, the greater likelihood that the original magnetization is overprinted by subsequent metamorphic or tectonic events, and the need for finding very precisely dated formations that can provide key poles in the construction of polarwander curves.

In this paper we report a new palaeomagnetic result for the Gawler Craton, South Australia, based on the very precisely dated Gawler Range Volcanics (Compston & Arriens, 1968; Webb, 1976). Comparison with results obtained previously (Chamalaun & Porath, 1968; Giddings & Embleton, 1976), suggests that the pole corresponding to 1500 m.y. is now well established for the Gawler Craton. With regard to the overall Australian Precambrian polarwander curve of McElhinny & Embleton (1976), the Gawler Craton data suggest only a minor modification in the age labelling. Furthermore, the results have placed a firm age on the final stages of the ore formation of the Middleback Ranges hematite ore bodies, and appear to favour a hypogene origin for the ore.

GEOLOGICAL SETTING

The Gawler Range Volcanics occur in the northern part of Eyre Peninsula in South Australia (see Fig. 1) and form the most extensive mass of calc-alkaline igneous rocks in Australia. Known outcrops cover some 25 000 km² but it is suspected that originally a much larger area may have been covered (Blissett, 1975). First discovered by Brown in 1885, the volcanics received little attention until Crawford (1963) published a description of large arc-like structures and Turner (1975) provided a detailed petrological account of the area near Uno. In 1972 the South Australian Department of Mines, in liaison with the Australian Mineral Development Laboratories, commenced a systematic program covering most of the area (Blissett, 1975, 1977).

The Gawler Range Volcanics consist predominantly of ignimbrites or welded ashflows. Petrologically they range from rhyolite through dacite to andesite and basalt. The basic members, however, make up only a small proportion of the total volcanic pile. The dominant

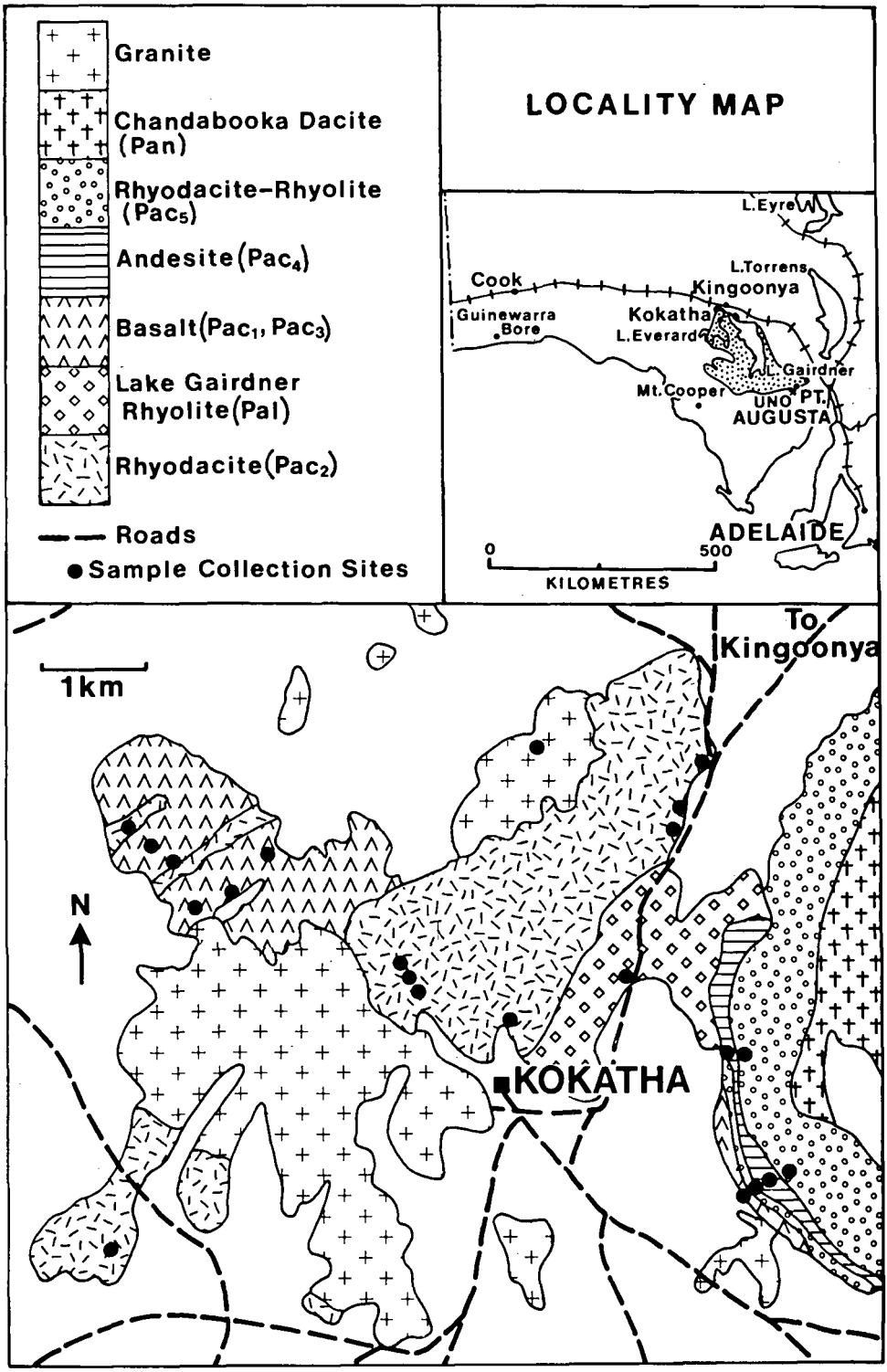


Fig. 1. Figure shows the palaeomagnetic sampling locations and main geological units near Kokatha homestead. Redrawn after Blissett (1975).

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rock types are dacite and rhyodacite, which are often porphyritic and at times show distinct flow banding.

Stratigraphically the Gawler Range Volcanics overlie the older Precambrian basement rocks of the Gawler Craton and, together with post-orogenic granites are in turn overlain in the east by later Precambrian and Lower Cambrian cover rocks. Thomson (1976) and Webb & Thomson (1977) concluded that the older Precambrian rocks of the Gawler Craton included relics of ancient (±2300 m.y.) basement complexes, that were overlain by the Lower Proterozoic Hutchison and Middleback Groups. Sedimentation was apparently terminated after *c.* 1900 m.y. and the early deep-seated initial phase of the Kimban orogeny was established *c.* 1800 m.y. Orogenesis involved deformation and synorogentic granite emplacement causing the Hutchison and Middleback Groups of sediments to be converted into the complexly deformed Cleve Metamorphics. The last phase of the Kimban orogeny commenced with the extrusion of the Gawler Range Volcanics, and terminated with the extrusion of postorogenic granites *c.* 1600-1450 m.y. This last plutonic event marked the final phase of cratonization of the Gawler Craton. Discussing the tectonic evolution of the Gawler Craton, Rutland (1973), and Glen *et al.* (1977), suggested that the craton was at a relatively high crustal level at the time of formation of the Gawler Range Volcanics and that its evolution may represent a Precambrian variant of a Cordilleran orogeny with subduction from the east. The Middleback Group contains banded iron formations which are host rocks to the hematite ore bodies, and for which a correlation with the Mt Bruce Supergroup in the Hamersley Ranges of Western Australia has been suggested (Thomson, 1976).

The Gawler Range Volcanics were originally dated by Compston *et al.* (1966) using a wide range of samples and yielded an age of 1505 ± 25 m.y.* More recently, further dating results were obtained by Webb (1976, 1977) on the Chitalanga Volcanics (1525 ± 15 m.y.*) and the overlying Chandabooka Dacite (1511 ± 36 m.y.*). The majority of the samples used in this study were obtained from the Chitanilga Volcanics and so we have adopted an age of 1525 m.y. as representative.

PALAEOMAGNETISM

Although the Gawler Range Volcanics cover a large area, only two areas have been described in sufficient detail for a palaeomagnetic study. Turner (1975) described the geology and petrology of the volcanics near Uno in the southeast. Blissett (1975) described the succession near Kokatha homestead in the north, where he defined various units and established the stratigraphical relationships. For this study we selected the Kokatha area as it contains within a relatively small area a good cross section of the Gawler Range Volcanics and in contrast to the Uno region contains good outcrops of the basalt and andesite members.

TABLE Summary of Palaeomagnetic Data.

N. Number of samples, DEC Declination, INC Inclination, α_{95} Radius of 95% confidence cone, K precision factor. Longitude of locality 135.3 E, Latitude 31.3 S.

	Before Magnetic Cleaning					After Magnetic Cleaning				
	N	DEC	INC	α_{95}	K	N	DEC	INC	α_{95}	K
Eastern Rhyolite - Rhyodacite (PAC_5)	11	27.0	-42.9	37	2.45	9	32.1	-42.3	11.6	20.70
Andesite (PAC_4)	8	358.9	-42.8	59	1.89	6	36.8	-64.0	11.3	33.62
Western Basalt (PAC_3)	19	34.2	-73.1	28	2.37	17	35.0	-47.4	8.5	18.79
Western Rhyodacite (PAC_2)	16	27.3	-39.8	29	2.49	12	34.1	-53.4	13.0	12.16
Eastern Basalt (PAC_1)	8	34.2	-52.3	9	36.97	8	34.2	-45.9	7.5	56.07
Mean all Samples						52	34.2	-48.1	5.2	15.07
Mean unit weight to groups						5	34.4	-50.1	6.5	109.21

* Ages thus marked here and elsewhere recomputed with ^{87}Rb half life of $1.42 \times 10^{-11} y^{-1}$.

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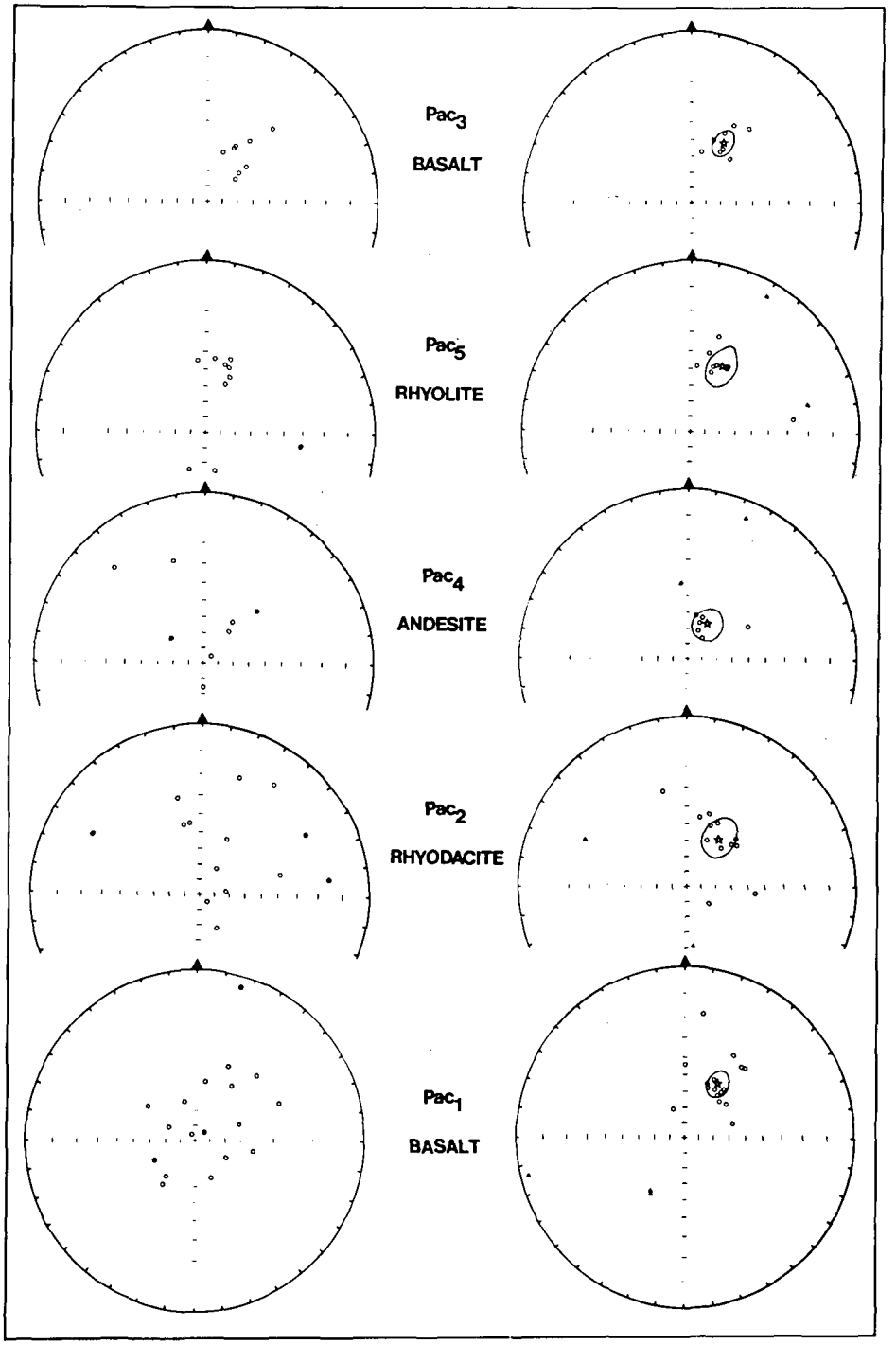
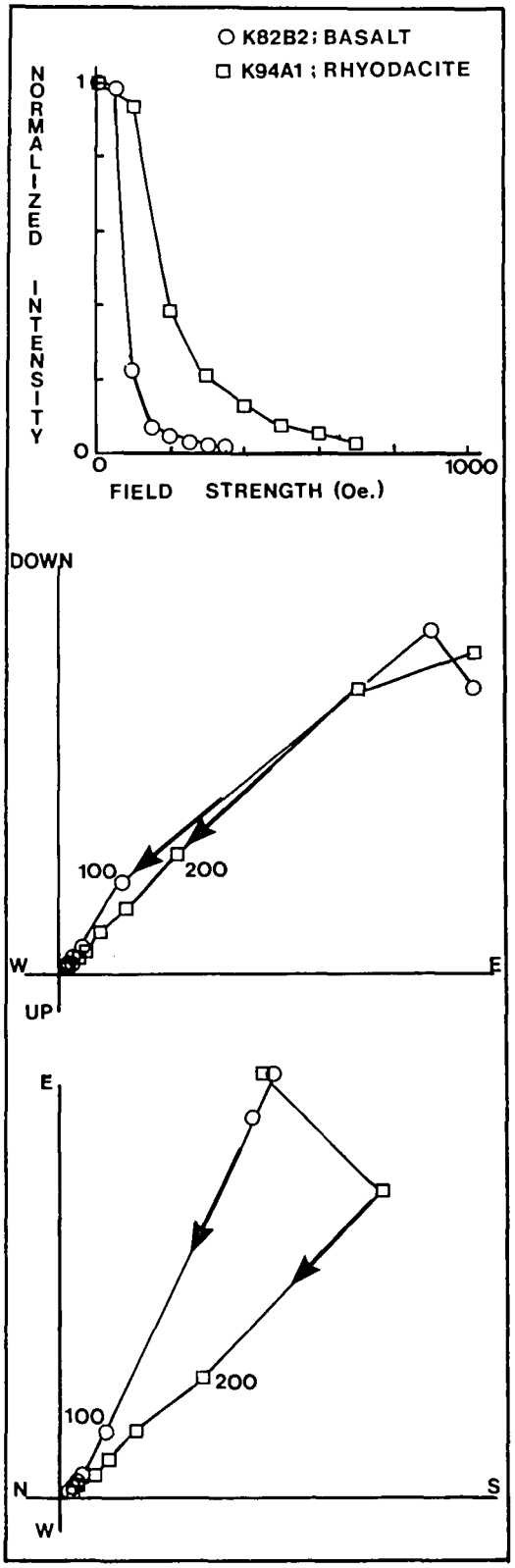


Fig. 2. Mean Sample directions of remanent magnetization before cleaning (left hand diagrams) and after magnetic cleaning (right hand diagrams), for the five principal rock units. Open symbols—negative inclinations. Closed symbols—positive inclinations. Triangles denote samples not included in the mean calculations. Stars indicate mean directions. Stereographic projection.

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Hand samples and drill cores were obtained, using a sun compass for orientation. Special care was exercised to obtain samples from deeply dissected creek beds to avoid weathering. A total of 62 samples were collected, providing a good cross section of the Gawler Range Volcanics. The remnant magnetization was measured with a Schonstedt spinner magnetometer.

Initially the collection was divided into five groups, based on stratigraphy and lithology. Sample localities are shown in Figure 1 and the main results summarized in the Table. The alphanumeric classification used in the Table is that proposed by Blissett (1975). The directions of the Natural Remanent Magnetization (N.R.M.) are shown in Figure 2 for each of the five groups. While the directions show considerable scatter the mean direction and general grouping is the same for all five groups. To improve the grouping and test for secondary components of magnetization, alternating field and thermal demagnetization techniques (As & Zijderveld, 1958; Irving *et al.*, 1961) were used. Initially one pilot specimen from each sample was demagnetized in a stepwise increasing alternating field. Figure 3 shows the change in directions and intensity as a function of A.F. field using the vector representation of As & Zijderveld (1958). The two specimens shown are fairly representative of the collection as a whole. The behaviour under A.F. demagnetization may be summarized as follows: The greatest change in direction occurs in relatively low fields of 100-200 oersted. After that the direction remains stable, until the intensity is reduced to a few percent of the original value. At that point the direction becomes random, presumably owing to spurious moments induced by the equipment. The two decay curves shown are extremes for the collection and suggest that magnetite is the carrier of the remanance in both the acid and basic volcanics. Whereas the basalt sample in Figure 3 shows a reduction of 10% of its original value at 100 oersted and the rhyodacite reaches that level at 400 oersted, the apparent correlation between lithology and magnetic hardness is, however, fortuitous. We have been unable to distinguish between acid and basic volcanics on the basis of their A.F. characteristics. This is a little surprising as the acid volcanics are often red,

Fig. 3. Example of behaviour during AF cleaning. Top diagram shows the decay of the (normalized) remanent magnetization as a function of AF field strength. Lower two diagrams show the change in direction. Orientation is arbitrary.

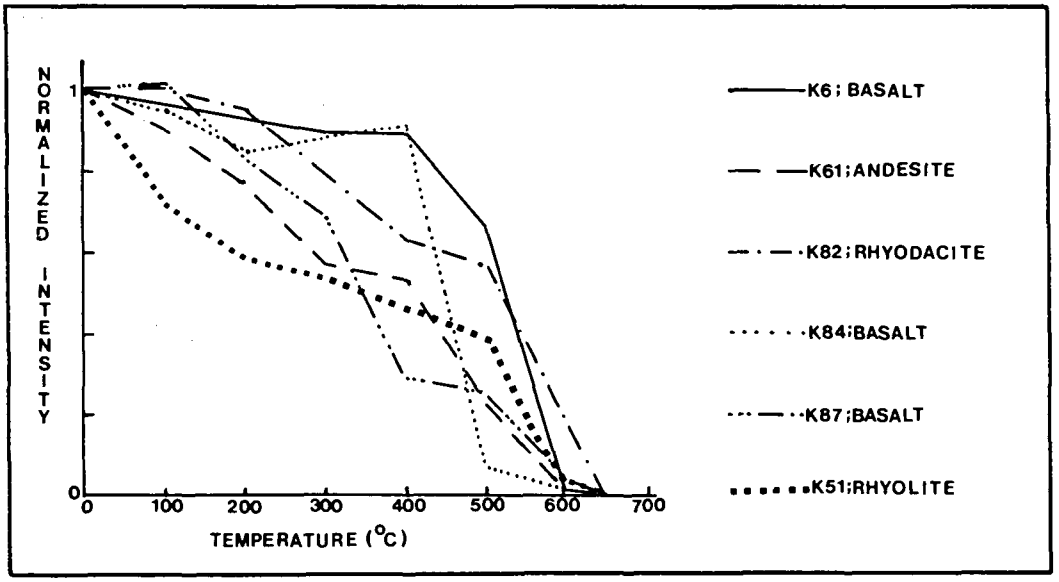


Fig. 4. Decay of the (normalized) remanent) magnetization as a function of temperature for selected lithologies.

suggestive of oxidizing conditions and hence could possibly contain a significant amount of hematite.

Selected specimens were also demagnetized using stepwise thermal demagnetization. The change in direction was the same as in A.F. demagnetization and unstable components were removed between 200-400°C. No consistent directions were obtained above 600°C. The thermal decay curves are shown in Figure 4 which confirm magnetite as the principal carrier of the remanence. X-ray analysis of the magnetic fraction separated from several samples covering the range of lithologies showed magnetite lines only. After examination of the stepwise demagnetization data for the pilot specimens, it was decided to clean the remainder of the collection in a field of 250 oersted, which was high enough to remove the unstable components, but not so high as to reduce the remanence to a level where spurious moments would mask the remaining stable remanence vector. The right-hand diagrams in Figure 2 show the cleaned sample directions and the relevant statistical parameters are included in the Table. The principal effect of A.F. cleaning has been to reduce the scatter in the directions of each of the five groups, but the mean directions did not change significantly. Apparently the unstable components were randomly directed. In calculating the mean direction for each group after cleaning, some odd directions were not included. It is not uncommon to find odd directions at a large angle from the main group. Following McElhinny *et al.* (1974), we have excluded those that made an angle larger than 60° from the mean. Excluding these odd directions does not affect the mean direction, but increases the dispersion factor a little. In a few cases the odd direction can be related to a suspected orientation error, but this does not apply to all samples and A.F. and thermal demagnetization data show that the odd samples are no different from the others in their magnetic properties and the direction is stable. Although we feel confident that the elimination procedure adopted is justified, the presence of such odd directions remains a matter of concern.

After cleaning, the mean directions of the individual groups are not significantly different (Table) and the dispersion of the individual groups small. It is commonly believed that ignimbritic volcanic piles are built up rapidly. The magma is highly gas charged and extruded in a short space of time. The palaeomagnetic results are consistent with that view and suggest that the individual units represent but a spot reading of the field with relatively little time difference between the units. In calculating a palaeomagnetic pole we have combined all the groups with the following result:

(a) giving unit weight to samples: Pole at 230 E, 60 N
(b) giving unit weight to each group: Pole at 233 E, 61 N.

As the difference is marginal we have chosen (a) as the representative pole.

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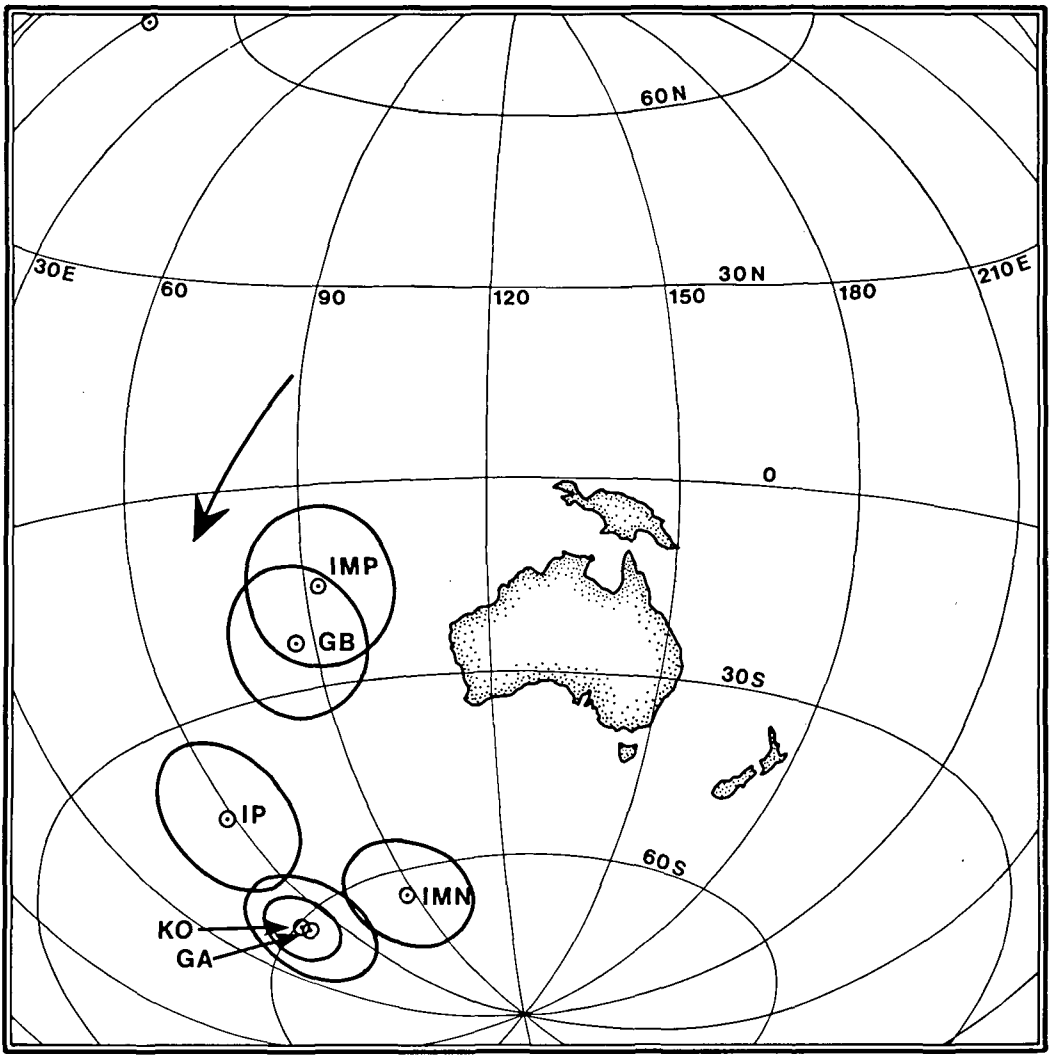


Fig. 5. Poles of the Gawler Craton. Arrow indicates direction of decreasing age. IMP Iron Monarch positive group, IMN Iron Monarch negative group, IP, Iron Prince negative group, GA negative dykes, GB positive dykes, KO Gawler Range Volcanics.

There is no evidence in the geological record that the region became involved in further deformational and metamorphic orogenies after extrusion of the Gawler Range Volcanics. The A.F. and thermal demagnetization experiments appear to have been successful in isolating a stable component, and one which is consistent between rocks of quite different lithologies. We regard the cleaned direction as the original and primary magnetization.

The results do not suggest that the palaeomagnetism is a useful tool for stratigraphy within the Gawler Range Volcanics themselves as no reversals were found and the difference in the directions of the different units is small indeed.

DISCUSSION

1. Gawler Craton palaeomagnetic poles

In addition to the Gawler Range Volcanics pole (Ko), there are five further poles available from the Gawler Craton from formatives of roughly the same age. They are shown in Figure 5, with associated (a_{95}) error circles.

Chamalaun & Porath (1968) investigated the palaeomagnetism of the hematite orebodies of the Middleback Ranges. Two orebodies, Iron Prince and Iron Monarch, yielded an acceptable palaeomagnetic result after A.F. and thermal cleaning. A single pole was presented for Iron Prince, which is designated IP in Figure 5. In the case of Iron Monarch two

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sets of directions were found: a positive group (IMp) and a negative group (IMn). These two groups of directions, while strongly suggestive of a geomagnetic polarity reversal, are not, however, antiparallel and were treated by Chamalaun & Porath (1968) as indicating two different orientations of the dipole axis. It was speculated that the ore-forming process may have covered a long enough time interval to have recorded a significant change in the dipole axis orientation. It was noted (Chamalaun & Porath, 1968) that there were no differences in the magnetic properties of the two Iron Monarch groups, nor were there obvious spatial, stratigraphic or petrological differences. The lack of antiparallelism between the two groups could have resulted from a secondary component having been added to an originally normal and reversed group. However, no evidence for such a secondary component was found in the A.F. and thermal demagnetization experiments.

A single pole was calculated for the Iron Prince Orebody. The majority of the directions were negative and the pole indicated a dipole axis orientation intermediate between IMn and IMp poles (see Fig. 5). Four samples were found with a positive inclination, but it could not be established statistically whether their mean direction should be treated as a separate group. Hence they were assumed to be true reversals by Chamalaun & Porah (1968). For the purpose of this paper, however, only the negative directions were used in the computation of the IP pole.

The age of the ore formation is difficult to establish. Chamalaun & Porath (1968) considered that the orebodies were probably younger than the metamorphic events of the Kimban orogeny, as there is no evidence to suggest that the hematite ore has been affected by the last major metamorphic event dated at 1745 ± 120 m.y.* by Compston & Arriens (1968). An upper limit may be placed on the ore-forming process, as the Corunna Conglomerate dated at 1505 ± 20 m.y.* (Compston *et al.*, 1966) does contain pebbles of iron ore (Miles, 1954).

Giddings & Embleton (1976) investigated the palaeomagnetism of a suite of basic dykes that intrude the older Precambrian basement complex in both Eyre and Yorke Peninsula. As in the case of Iron Monarch two sets of directions were obtained. One group of dykes (GA) showed negative inclinations and the other (GB) positive inclinations. As the two groups are not strictly antiparallel, separate poles were calculated, as in the case of the Iron Monarch results. Apart from the difference in the direction of magnetization, no difference in petrology, dyke orientation, stratigraphy, or other magnetic properties were noted for the two sets of dykes. Giddings & Embleton (1976) also discount any selective remagnetization or regional thermal event, as a possible explanation for the lack of antiparallelism between the two groups. They quote an age of 1700 ± 100 m.y. for the GB dykes and 1500 ± 200 m.y. for the GA dykes, but no further details are provided.

Figure 5 shows that the five data sets are in excellent agreement. The poles derived from the negative directions Ko, IMn and GA are in agreement with significant overlap of the error circles and the positive direction IMp and GB are similarly in agreement. The Iron Prince negative group occupies an intermediate position. The agreement between Ko, IMn and GA firmly establishes the reliability of these poles. The possibility that they are present day directions, already regarded as unlikely (Giddings & Embleton, 1976) must now be thoroughly discounted. It is most unlikely that three quite different and widely separated formations acquired the same component of secondary magnetization, which was equally resistant to magnetic cleaning. Of the three formations the Gawler Range Volcanics is the most accurately dated information at 1525 ± 15 m.y. and we suggest that the Ko pole be regarded as the key pole for that age.

The agreement between the poles derived from the positive directions (IMp and GB) by itself suggests that the orientation of the dipole axis implied by these data is real. The possibility that the positive results represent selective remagnetization involving a secondary component superimposed on an exact reversed polarity must be ruled out. If the reliability of the negative directions is accepted, then, since no magnetic or other differences have been observed between the positive and negative groups of the same formations, the positive results should be regarded as equally reliable.

2. Implication for the hematite orebodies

The origin of the Middleback Ranges hematite orebodies is not resolved (Furber & Cook, 1976). It is generally accepted that the ore was formed from banded iron formations by the removal of silica and the introduction of additional iron. Magnetic anisotropy studies (Chamalaun & Porath, 1968) confirmed geological observations which suggested that the original sedimentary banding has been preserved in the ore. The two basic hypotheses proposed are: a hypogene origin in which hot

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fluids, possibly magmatic in origin caused the replacement, or a supergene origin involving relative cold waters and subsurface leaching. Miles (1954) argued for a hypogene origin, whereas Whitehead (1962), Ashworth (1965), and Furber & Cook (1976) favour a supergene origin. The latter authors, however, note that many of the objections to a supergene origin remain unanswered.

The palaeomagnetic agreement between the positive Iron Monarch directions (IMp) and the positive GB dykes on the one hand, and the negative Iron Monarch (IMn) and GA dyke directions on the other, could imply that they are contemporaneous and possibly related genetically. It may be speculated that the ore-forming process began with a general upwarping of the craton, causing rifting, and the injection of dykes. The ore began to be formed with the injection of GB dykes, and is now recognised by its positive directions. It was completed with the injection of the GA dykes and the ore then formed acquired negative directions. In this model the close association between the dykes and the ore must be regarded as strong argument for a hypogene origin. Evidently the ore was formed at a time that igneous activity was present and hydrothermal fluids could have been injected to penetrate along structural weak zones together with the dykes.

These considerations apply mainly to the Iron Monarch and Iron Prince orebodies. It is not clear whether they should be applied to all the orebodies. Thus, for example, the Iron Duke orebody appears to be different in a number of respects, but primarily in containing carbonate ore at depth. It is possible therefore that its origin, while perhaps still triggered by igneous activity, does involve a different mechanism.

The model suggests that the last phase of ore formation is dated by the Gawler Range Volcanics at close to 1525 m.y. The starting age of the ore-forming process is less precisely estimated. In the model it is equated with the age of the GB dykes, but their age has an analytical uncertainty of $\pm$ 100 m.y. The youngest estimate would be 1600 m.y. suggesting an overall time for the ore-forming process of 100 m.y., from 1600 m.y. to 1500 m.y.

The first recorded phase of igneous activity that culminated in the extrusion of the Gawler Range Volcanics, appears to have the intrusion of the Tarcoola granites at about 1546 m.y.* (Glen *et al.*, 1977; Thomson, 1976). Thus it seems reasonable to suppose that the iron ore development and dyke intrusions are associated with and a manifestation of that episode of igneous activity. However, the intriguing question that arises is whether the iron ore forming process was more or less continuous during this 100 m.y. period, or whether it commenced with the intrusion of the GB dykes, paused for a while and restarted when the GA dykes were intruded. The directions of magnetization of the Iron Monarch body are scattered, but not strongly streaked, as might have been expected for a continuous ore-forming process. The suggestion is, therefore, in favour of two pulses of ore formation. It implies, however, that the crustal weaknesses that provided access to the ore for the hydrothermal suite remained much the same over a 100 m.y. period. This is perhaps surprising, but the apparent similarity in the orientation of the two dyke suites does suggest that the stress field was roughly similar.

The alternative hypothesis is that the two dyke suites are not in fact significantly different in age, and that the whole process of dyke intrusion and ore formation was completed within a relative short time interval (perhaps 30 m.y. or so) just prior to the extrusion of the Gawler Range Volcanics. The main difficulty with this alternative is, however, that the palaeomagnetic results have to be interpreted as indicating an unusual rapid rate of polar wander of 2° to 3° per million years.

We prefer the two-pulse ore-forming process as it is more consistent with the geological and palaeomagnetic constraints. However, a more detailed study of the petrology, age and structural relationship of the dyke suites would help to evaluate the alternative hypotheses.

3. Precambrian polarwander curve

McElhinny & Embleton (1976) have recently presented a summary of Precambrian palaeomagnetic poles for Australia and drawn up an apparent polarwander curver from *c.* 2500 m.y. to 650 m.y. Although there are as yet few pole pairs from different cratons of the same age, they considered that a single polarwander swath (McElhinny & McWilliams. 1977) could be drawn through all the data.

The main effect of the result of the Gawler Range Volcanics is to provide a key pole at 1500 m.y. Originally McElhinny & Embleton had labelled the curve at 1600 m.y. corresponding to the GA pole, but in view of the new Ko pole it would now be more appropriate to associate it with a 1500 m.y. label. The GB pole and IMp pole are now on the basis of the discussion in the previous section more appropriately labelled at 1600 m.y. The rate of polarwander indicated by this segment is about 0.4°/m.y. which is not much different from

the average for the Phanerozoic (McElhinny, 1973), but this depends on the nature of the ore-forming process as discussed above.

It is of interest to note that assuming a constant rate of polarwander for the interval 1600 m.y. to 1500 m.y. the polarity reversal appears to have occurred at about 1550 m.y. almost coincident with the formation of the IP orebody. The polarity indicated by the Mt Goldsworthy Ore (MG2) (Porath & Chamalaun), 1968) and YD dykes of Western Australia (Giddings, 1976) is consistent with that interpretation. Since this (Gawler) reversal appears to separate two relatively long periods of constant polarity, it may be a useful magnetic stratigraphical horizon in the Precambrian.

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